Singapore School Kinderland

Continual Assessment 1

2026 - 2027

Secondary 2

Computer Science

Part 1

Date: _____ Name: _____

Instructions:

- Start answering the questions on time.
- Follow all instructions carefully.
- Answer all questions.
- Submit your paper within the given time.

Total marks	**35**
Part 1	**30**
CW	**5**
Marks Obtained	

Section A: Multiple Choice Questions (MCQ) (8x1=8 marks)

For each question 1-8, four options are given. Circle the correct option i, ii, iii or iv.

1. Which number system uses Base-16 and the digits 0–9 along with letters A–F?

i. Binary **ii.** Denary
iii. Hexadecimal **iv.** Octal

2. How many bits make up exactly 1 Byte of digital data?

i. 4 bits **ii.** 8 bits
iii. 16 bits **iv.** 1,024 bits

3. How many bits are used by the standard ASCII character set to represent one character?

i. 7 bits **ii.** 16 bits
iii. 32 bits **iv.** 64 bits

4. What is the standard unit used to measure the Sample Rate of a digital sound wave?

i. Pixels per inch (PPI) **ii.** Bytes per second
iii. Hertz (Hz) **iv.** Bits per pixel

5. Which type of file compression reduces file size WITHOUT losing any original data?

i. Lossy compression **ii.** Lossless compression
iii. Analogue compression **iv.** Pixel compression

6. In the IEC binary storage standard, how many Bytes equal 1 Kibibyte (1 KiB)?

i. 100 Bytes **ii.** 512 Bytes
iii. 1,000 Bytes **iv.** 1,024 Bytes

7. If a bitmap image has a colour depth of 8 bits per pixel, how many colours can it display?

i. 8 colours **ii.** 16 colours
iii. 128 colours **iv.** 256 colours

8. Which hardware component converts digital binary audio back into analogue sound for speakers?

i. ADC **ii.** DAC
iii. MAC **iv.** RLE

Section B: True or False (6x1=6 marks)

9. Write True or False in the box provided for each statement:

No.	Statement	True / False
i.	Computers use the Binary (Base-2) system because switches have two states: ON (1) and OFF (0).	
ii.	An image with a 4-bit colour depth can display a maximum of 16 different colours.	
iii.	Decreasing the sample rate and sample resolution of an audio file makes sound quality better.	
iv.	Lossy compression permanently removes unnecessary data that humans cannot easily notice.	
v.	Standard 7-bit ASCII can represent every language in the world including Chinese and emojis.	
vi.	Higher image resolution means more pixels are used, which increases the image file size.	

Section C: Fill in the Blanks (6x1=6 marks)

10. Fill in the blanks using the correct words from the box below:

Word Bank: [Pixel | ADC | Unicode | MAC Address | Metadata | Resolution]

i. A _____ is the smallest single dot or picture element that makes up a bitmap image.

ii. The _____ character set uses 16 to 32 bits to represent characters from all world languages.

iii. An _____ (Analogue-to-Digital Converter) converts analogue sound waves into digital binary data.

iv. Image _____ stores extra file details such as image width, height, and colour depth.

v. A _____ is a unique hardware identifier written in Hexadecimal (e.g., 00-1C-B3-4F-25-FE).

vi. Image _____ refers to the total number of pixels (Width × Height) in a digital image.

Section D: Number Systems & Hexadecimal (3+2=5 marks)

11. Complete the conversions below (3m) and state two reasons for using Hexadecimal (2m):

No.	Conversion Question	Final Answer
i.	Convert 8-bit Binary **1010 1110** to Denary:	
ii.	Convert 12-bit Binary **0111 1101 0100** to Hexadecimal:	
iii.	Convert Hexadecimal **5B8** to 12-bit Binary:	

iv. Reason 1:

_____

v. Reason 2:

_____

Section E: Data Compression & File Size (3+1+1=5 marks)

12. Tick (✓) Lossy or Lossless below (3m), and answer (iv) RLE & (v) File Size (2m):

No.	File Type / Scenario	Lossy (✓)	Lossless (✓)
i.	A photograph saved as a **JPEG** file for a website.		
ii.	A computer software program file (**program.exe**).		
iii.	An **MP3** music file used for online streaming.		

iv. RLE Compression of [**W W W W W W B B B W W W W B B B**]:
_____ **(1m)**

v. File Size of **4,096 characters** (1 Byte each) in **KiB** (1 KiB = 1,024 Bytes): _____ **(1m)**